Flora Care Manager: Smart Console for Plant Care

> A C++-based interactive system designed for meticulous management, tracking, and recording of care details for diverse plant collections.

Robust C++ Implementation

Leverages Object-Oriented Programming (OOP) principles for flexible and modular design.

Interactive Monitoring

Enables easy registration, dynamic monitoring, and precise tracking of watering schedules.

Persistent Storage

Utilizes file operations for reliable and persistent storage of home or greenhouse collection data.

Motivation & The Real-World Gap in Plant Care

Tracking Tedium

Many plant owners lack a simple, unified tool to track diverse water schedules and unique care needs for various plant types.

Error Prone

Most basic logging methods—journals, sticky notes, or memory—are inefficient, tedious, and highly prone to error or loss.

Project Goal

Design a lightweight, user-friendly console application to professionally organize and manage an entire plant collection in one centralized place.

Core Features: Key Capabilities at a Glance

Customizable Plant Types

Registers different categories (Succulent, Flower, Cactus, Water Plant) utilizing powerful OOP inheritance for unique attribute handling.

Reliable Data Persistence

Stores and reloads all collected data reliably across runs using robust plain-text file Input/Output (I/O).

Interactive Console System

Features a menu-driven command structure for easy adding, viewing, watering, and updating of plant records.

Zero Memory Leaks

Guarantees comprehensive memory cleanup on program exit, demonstrating efficient dynamic memory management practices.

The system uses polymorphism to treat all plant types uniformly while maintaining their specific properties, simplifying the management code.

Technical Architecture: C++ Code Structure

The Flora Care Manager is built on a strong object-oriented foundation, ensuring modularity and scalability.

Core Class Hierarchy

Plant serves as the abstract base class. Derived classes—Succulent, Flower, Cactus, and WaterPlant—inherit and extend common features.

Data Initialization

The system begins by attempting to load and deserialize all existing saved plant data from the persistence file.

Main Program Loop

An interactive menu loop handles user commands: Add, View, Water, and Exit. User input drives the core functionality.

Persistence & Clean Exit

Upon receiving the exit command, all current plant records are serialized and saved back to the file, followed by a clean memory deallocation.

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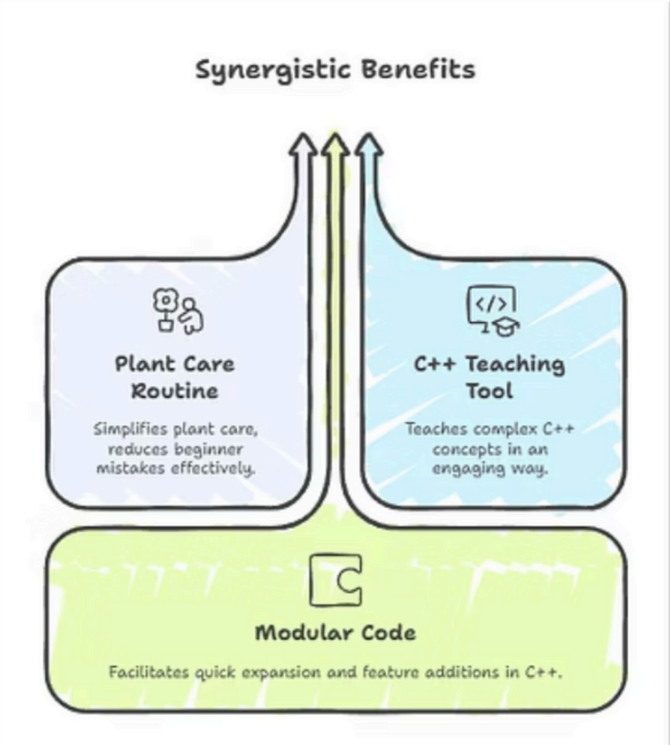
Real-World Impact & Future Ideas

Immediate Benefits

- Makes plant care routine and reliable, reducing beginner error and aiding enthusiasts.
- Serves as an excellent teaching tool for complex C++ concepts (OOP, dynamic memory, I/O).
- The modular C++ code is ready for quick expansion and feature additions.



Potential Future Extensions

Mobile App Integration

Expand console functionality with a user-friendly mobile interface for on-the-go access.

Automated Notifications

Implement timed reminders for critical events like watering, feeding, or repotting.

Visual Analytics

Integrate charts and graphs to visualize care history, growth rates, and health trends.

Thank You

ES203 - OOPs using C++ MINI PROJECT

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This project demonstrates that complex programming principles can be applied to solve simple, practical problems with elegance and efficiency.